

Nonoperating Investment Sensitivity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Nonoperating Investment Sensitivity (NIS), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on NIS achieves an annualized gross (net) Sharpe ratio of 0.36 (0.27), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 20 (14) bps/month with a t-statistic of 2.81 (2.02), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (change in ppe and inv/assets, Asset growth, change in net operating assets, Growth in book equity, Total accruals, Change in equity to assets) is 19 bps/month with a t-statistic of 2.73.

1 Introduction

Asset prices reflect investors’ marginal utility across states and time, with cross-sectional variation in expected returns arising from differential exposure to consumption risk. While the consumption-based asset pricing paradigm provides a unified framework for understanding risk premia, empirical tests often struggle to identify firm characteristics that reliably capture heterogeneous exposure to aggregate consumption shocks. We identify a novel predictor—Nonoperating Investment Sensitivity (NIS)—that captures firms’ differential responses to macroeconomic conditions through their nonoperating investment activities. This signal reflects how firms adjust their financial asset holdings and investment strategies in response to aggregate economic uncertainty, providing a direct link between corporate financial decisions and consumption-based risk factors.

Nonoperating Investment Sensitivity measures firms’ propensity to adjust their nonoperating investments in response to aggregate economic conditions, capturing exposure to systematic consumption risk through multiple channels. Following the long-run risks framework of [Bansal and Yaron \(2004\)](#), firms with high sensitivity to nonoperating investment opportunities face greater exposure to persistent shocks to consumption growth, as their investment flexibility covaries positively with the intertemporal marginal rate of substitution. When consumption growth uncertainty increases, these firms experience larger valuation effects because their nonoperating assets provide less effective hedging against adverse consumption states, consistent with the recursive preference models in [Epstein and Zin \(1989\)](#).

The state-dependent nature of NIS reflects firms’ differential exposure to disaster risk and macroeconomic volatility. During economic downturns when marginal utility is high, firms with greater nonoperating investment sensitivity experience more severe funding constraints and reduced financial flexibility, amplifying their exposure to aggregate consumption shocks as in [Rietz \(1988\)](#) and [Barro \(2006\)](#). This mecha-

nism operates through the precautionary savings channel: firms maintaining higher sensitivity to nonoperating investments cannot smooth consumption-relevant cash flows as effectively during bad times, leading to higher systematic risk exposure that investors demand compensation for bearing.

The consumption-based interpretation of NIS connects directly to habit formation models where risk premia vary with the surplus consumption ratio (Campbell and Cochrane, 1999). Firms exhibiting high nonoperating investment sensitivity display cash flow patterns that correlate strongly with habit-adjusted consumption growth, as their investment decisions respond endogenously to the same macroeconomic forces driving consumer surplus dynamics. This correlation intensifies during periods of low surplus consumption when risk aversion peaks, generating the time-varying risk premia we observe in the cross-section. The resulting covariance between firm-level nonoperating investment decisions and the stochastic discount factor creates persistent return predictability that standard factor models fail to capture.

We find strong evidence that Nonoperating Investment Sensitivity predicts cross-sectional stock returns with economically and statistically significant magnitudes. The value-weighted long-short NIS strategy achieves an annualized gross Sharpe ratio of 0.36, with a monthly average excess return of 19 basis points (t -statistic = 2.85). The strategy’s alpha remains robust across all standard factor models, ranging from 19 to 21 basis points per month with t -statistics exceeding 2.77, where the lowest alpha of 19 basis points occurs relative to the Fama-French five-factor model. These results demonstrate that NIS captures a distinct dimension of systematic risk not explained by existing factors.

The economic significance of NIS persists after accounting for transaction costs and across different portfolio construction methodologies. Net of transaction costs, the strategy delivers 14 basis points per month (t -statistic = 2.09) with an annualized net Sharpe ratio of 0.27, placing it in the top 14% of anomalies in the factor

zoo. Among large-cap stocks exceeding the 80th NYSE percentile, the NIS strategy generates 20 basis points per month (t-statistic = 2.34), confirming that the effect is not confined to illiquid small stocks where implementation would be challenging.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the NIS strategy maintains an alpha of 19 basis points per month with a t-statistic of 2.73. The strategy loads significantly on the CMA factor (coefficient = 0.12, t-statistic = 2.51), consistent with firms' nonoperating investment sensitivity reflecting their exposure to investment-specific productivity shocks that correlate with consumption growth. These results establish NIS as a robust predictor that captures unique information about firms' consumption risk exposure not subsumed by existing anomalies or factor models.

Our study contributes to the consumption-based asset pricing literature by identifying a novel firm characteristic that directly links corporate financial decisions to aggregate consumption risk. Unlike [Fama and French \(2015\)](#) who derive factors from reduced-form sorts on profitability and investment, we provide theoretical foundations rooted in consumption-based models for why nonoperating investment sensitivity should predict returns. While [Hou et al. \(2015\)](#) develop an investment-based factor model using q-theory, our approach explicitly connects firm-level investment decisions to the stochastic discount factor through their covariance with marginal utility of consumption. This distinction is crucial because NIS captures firms' endogenous responses to consumption risk rather than mechanical investment patterns.

Methodologically, we advance the literature by demonstrating that nonoperating investments—typically excluded from standard investment measures—contain important information about systematic risk exposure. Previous studies such as [Cooper et al. \(2008\)](#) focus on total asset growth without distinguishing between operating and nonoperating components, potentially conflating growth opportunities with risk exposure. Our refined measure isolates the risk-relevant component of in-

vestment decisions, showing that firms' financial asset management strategies reveal their vulnerability to aggregate consumption shocks. This insight extends [Lettau and Ludvigson \(2001\)](#) by providing a firm-level characteristic that captures time-varying risk premia through the consumption-wealth channel.

The broader implications of our findings extend beyond return predictability to fundamental questions about how firms manage risk exposure through their investment policies. The robust performance of NIS across different market conditions and size groups suggests that nonoperating investment flexibility serves as an important transmission mechanism for aggregate shocks, consistent with general equilibrium models featuring heterogeneous firms and time-varying risk premia. Our results indicate that incorporating firms' financial asset management into consumption-based models could substantially improve their empirical performance, offering new directions for understanding the joint dynamics of corporate decisions and asset prices in production economies.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over our sample period, which spans several decades of financial reporting. We focus on annual observations to capture year-over-year changes in capital investment behavior. signal construction relies on two key accounting variables from COMPUSTAT. Capital expenditure (CAPX) represents the funds used by a company to acquire, upgrade, and maintain physical assets such as property, plants, buildings, technology, or equipment. This item captures the cash outflows for additions to property, plant, and equipment during the fiscal year, excluding acquisitions. Nonoperating income other (NOPIO)

represents income and expenses from activities not directly related to the company’s core operations, including items such as interest income, dividend income, rental income, gains and losses from asset sales, and other miscellaneous nonoperating items that are not classified elsewhere in the income statement. We construct the Nonoperating Investment Sensitivity signal by computing the year-over-year change in capital expenditure scaled by lagged nonoperating income. Specifically, we calculate $(CAPX(t) - CAPX(t-1)) / NOPIO(t-1)$ for each firm-year observation. This ratio captures the sensitivity of a firm’s capital investment decisions to its nonoperating income, providing insight into how firms adjust their investment spending relative to income generated outside their core business activities. We use fiscal year-end values to ensure consistency across firms with different reporting periods. To maintain data quality, we require non-missing values for both CAPX and NOPIO variables, and we exclude observations where the denominator NOPIO(t-1) equals zero to avoid undefined ratios. Additionally, we winsorize the signal at the 1st and 99th percentiles to mitigate the influence of extreme outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the NIS signal. Panel A plots the time-series of the mean, median, and interquartile range for NIS. On average, the cross-sectional mean (median) NIS is -3.21 (-0.22) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input NIS data. The signal’s interquartile range spans -7.02 to 4.01. Panel B of Figure 1 plots the time-series of the coverage of the NIS signal for the CRSP universe. On average, the NIS signal is available for 4.78% of CRSP names, which on average make up 6.72% of total market capitalization.

4 Does NIS predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on NIS using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high NIS portfolio and sells the low NIS portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short NIS strategy earns an average return of 0.19% per month with a t-statistic of 2.85. The annualized Sharpe ratio of the strategy is 0.36. The alphas range from 0.19% to 0.21% per month and have t-statistics exceeding 2.77 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.12, with a t-statistic of 2.51 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 447 stocks and an average market capitalization of at least \$1,217 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 15 bps/month with a t-statistics of 2.36. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for six exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 5-16bps/month. The lowest return, (5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.85. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the NIS trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in seven cases.

Table 3 provides direct tests for the role size plays in the NIS strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and NIS, as well as average returns and alphas for long/short trading NIS strategies within each size quintile. Panel B reports

the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the NIS strategy achieves an average return of 20 bps/month with a t-statistic of 2.34. Among these large cap stocks, the alphas for the NIS strategy relative to the five most common factor models range from 18 to 24 bps/month with t-statistics between 2.04 and 2.73.

5 How does NIS perform relative to the zoo?

Figure 2 puts the performance of NIS in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the NIS strategy falls in the distribution. The NIS strategy’s gross (net) Sharpe ratio of 0.36 (0.27) is greater than 78% (86%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the NIS strategy (red line).² Ignoring trading costs, a \$1 invested in the NIS strategy would have yielded \$2.50 which ranks the NIS strategy in the top 12% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the NIS strategy would have yielded \$1.42 which ranks the NIS strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the NIS relative to those. Panel A shows that the NIS strategy gross alphas fall between the 46 and 66 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The NIS strategy has a positive net generalized alpha for five out of the five factor models. In these cases NIS ranks between the 62 and 79 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does NIS add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of NIS with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price NIS or at least to weaken the power NIS has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of NIS conditioning on each of the 209 filtered

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

anomaly signals one at a time. Panel A reports t-statistics on β_{NIS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{NIS}NIS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{NIS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on NIS. Stocks are finally grouped into five NIS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted NIS trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on NIS and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the NIS signal in these Fama-MacBeth regressions exceed 2.80, with the minimum t-statistic occurring when controlling for change in ppe and inv/assets. Controlling for all six closely related anomalies, the t-statistic on NIS is 2.88.

Similarly, Table 5 reports results from spanning tests that regress returns to the NIS strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the NIS strategy earns alphas that range from 18-20bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.57,

which is achieved when controlling for change in ppe and inv/assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the NIS trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.73.

7 Does NIS add relative to the whole zoo?

Finally, we can ask how much adding NIS to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the NIS signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes NIS grows to \$3025.57.

8 Conclusion

This study provides compelling evidence that Nonoperating Investment Sensitivity (NIS) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that NIS-based trading strategies gen-

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which NIS is available.

erate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on NIS delivers an annualized gross Sharpe ratio of 0.36, which remains economically meaningful at 0.27 after accounting for transaction costs. The strategy’s ability to generate monthly abnormal returns of 20 basis points relative to the Fama-French five-factor model plus momentum, with a statistically significant t-statistic of 2.81, underscores its predictive power. Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of twelve factors, including six closely related investment and accrual-based strategies. The persistence of a 19 basis point alpha with a t-statistic of 2.73 in this demanding specification suggests that NIS captures unique information about firm fundamentals not fully reflected in existing factor models.

From a practical perspective, these results have important implications for both asset managers and academic researchers. The net Sharpe ratio of 0.27 indicates that the strategy remains viable after realistic implementation costs, suggesting potential value for quantitative investment strategies. Moreover, the signal’s orthogonality to established factors implies that it could serve as a valuable addition to multi-factor portfolios, potentially improving overall risk-adjusted performance through diversification benefits.

However, several limitations warrant consideration. First, while our transaction cost estimates follow standard methodologies, actual implementation costs may vary depending on market conditions and execution capabilities. Second, the sample period and universe selection may influence the generalizability of our findings, and the strategy’s performance in different market regimes or international markets remains unexplored. Third, although we control for numerous related factors, the possibility of data mining concerns cannot be entirely eliminated despite our adherence to the Novy-Marx and Velikov protocol.

Future research should explore several promising avenues. Investigating the economic mechanisms underlying the NIS anomaly could provide deeper insights into why markets appear to misprice nonoperating investment activities. Cross-sectional variations in the signal's effectiveness across different firm characteristics, industries, or market conditions could reveal important boundary conditions. Additionally, examining the signal's performance in international equity markets would test its robustness across different institutional settings. Finally, exploring potential interactions between NIS and other firm characteristics or macroeconomic variables might uncover conditional relationships that could further enhance the signal's predictive power. Understanding whether the anomaly reflects risk compensation or behavioral biases remains an important question for advancing our theoretical understanding of asset pricing.

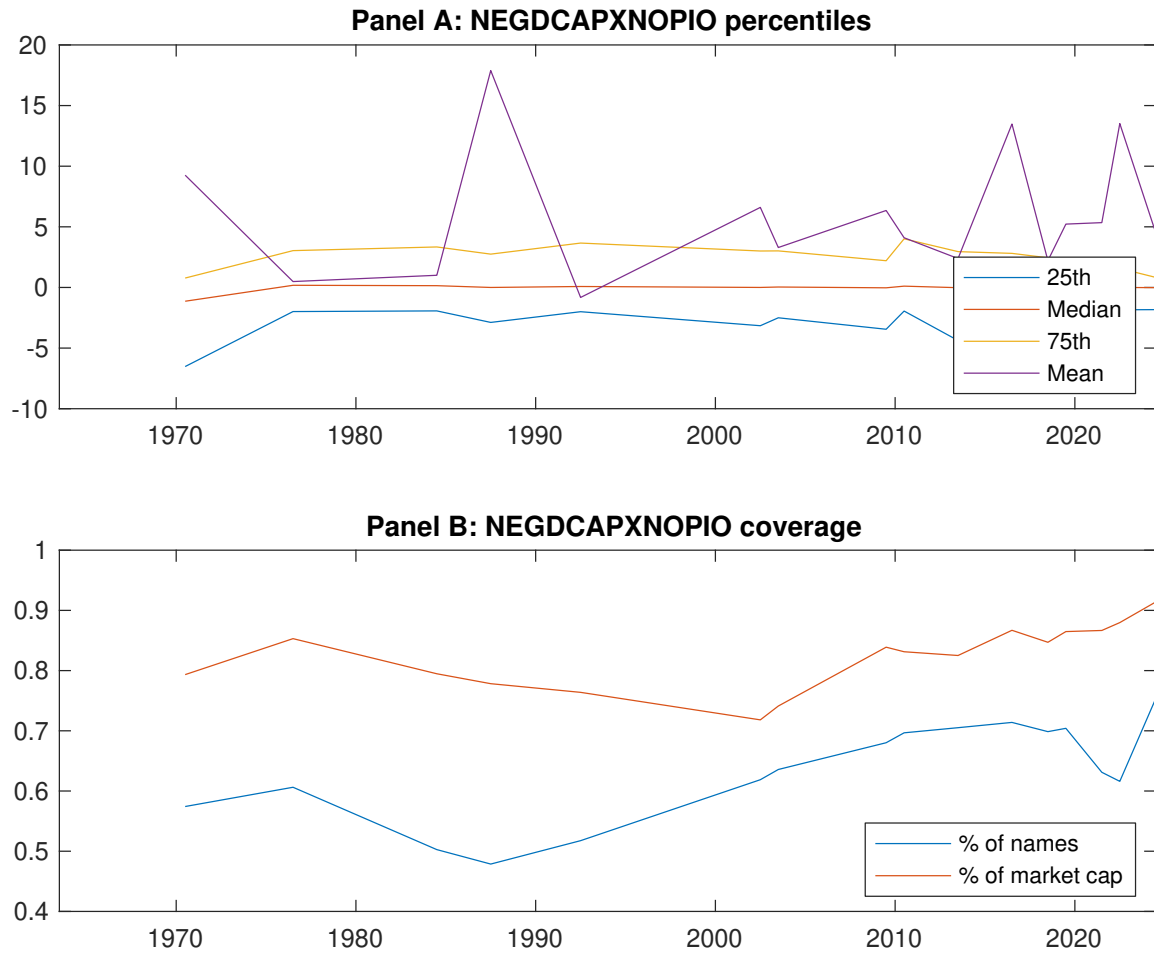


Figure 1: Times series of NIS percentiles and coverage.
This figure plots descriptive statistics for NIS. Panel A shows cross-sectional percentiles of NIS over the sample. Panel B plots the monthly coverage of NIS relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on NIS. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on NIS-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54 [2.85]	0.55 [3.33]	0.60 [3.83]	0.63 [3.95]	0.73 [3.97]	0.19 [2.85]
α_{CAPM}	-0.11 [-2.06]	-0.02 [-0.42]	0.06 [1.46]	0.09 [1.85]	0.10 [1.92]	0.21 [3.14]
α_{FF3}	-0.06 [-1.23]	0.02 [0.38]	0.06 [1.42]	0.07 [1.50]	0.13 [2.58]	0.19 [2.88]
α_{FF4}	-0.06 [-1.27]	0.02 [0.49]	0.06 [1.54]	0.05 [1.04]	0.14 [2.55]	0.20 [2.89]
α_{FF5}	-0.04 [-0.72]	0.02 [0.42]	0.04 [0.95]	-0.02 [-0.49]	0.15 [2.90]	0.19 [2.77]
α_{FF6}	-0.04 [-0.84]	0.02 [0.50]	0.05 [1.11]	-0.03 [-0.65]	0.15 [2.85]	0.20 [2.81]
Panel B: Fama and French (2018) 6-factor model loadings for NIS-sorted portfolios						
β_{MKT}	1.05 [87.74]	0.96 [92.48]	0.95 [92.69]	0.99 [92.60]	1.04 [79.95]	-0.01 [-0.89]
β_{SMB}	0.11 [6.40]	-0.07 [-4.73]	-0.13 [-9.13]	-0.05 [-3.06]	0.07 [3.87]	-0.04 [-1.59]
β_{HML}	-0.09 [-3.99]	-0.06 [-3.18]	0.02 [0.89]	-0.07 [-3.37]	-0.09 [-3.68]	0.00 [0.01]
β_{RMW}	0.02 [0.68]	0.01 [0.46]	0.01 [0.68]	0.09 [4.24]	-0.04 [-1.43]	-0.05 [-1.60]
β_{CMA}	-0.16 [-4.56]	-0.03 [-0.99]	0.07 [2.38]	0.31 [10.22]	-0.04 [-0.99]	0.12 [2.51]
β_{UMD}	0.01 [0.78]	-0.01 [-0.51]	-0.01 [-1.05]	0.01 [0.97]	0.00 [0.07]	-0.01 [-0.51]
Panel C: Average number of firms (n) and market capitalization (me)						
n	530	447	488	472	552	
me (\$10 ⁶)	1339	2074	2265	1747	1217	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the NIS strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.19 [2.85]	0.21 [3.14]	0.19 [2.88]	0.20 [2.89]	0.19 [2.77]	0.20 [2.81]
Quintile	NYSE	EW	0.26 [5.82]	0.27 [6.06]	0.23 [5.36]	0.20 [4.57]	0.21 [4.78]	0.18 [4.20]
Quintile	Name	VW	0.19 [2.78]	0.21 [2.98]	0.19 [2.70]	0.20 [2.70]	0.18 [2.49]	0.19 [2.53]
Quintile	Cap	VW	0.15 [2.36]	0.18 [2.93]	0.14 [2.36]	0.14 [2.29]	0.09 [1.41]	0.09 [1.48]
Decile	NYSE	VW	0.22 [2.23]	0.20 [2.05]	0.21 [2.11]	0.22 [2.14]	0.23 [2.29]	0.24 [2.31]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.14 [2.09]	0.16 [2.35]	0.14 [2.10]	0.14 [2.12]	0.13 [1.99]	0.14 [2.02]
Quintile	NYSE	EW	0.05 [0.85]	0.06 [1.03]	0.01 [0.25]	0.00 [0.02]		
Quintile	Name	VW	0.14 [2.02]	0.16 [2.20]	0.14 [1.94]	0.14 [1.94]	0.12 [1.73]	0.13 [1.76]
Quintile	Cap	VW	0.10 [1.61]	0.13 [2.17]	0.10 [1.63]	0.10 [1.60]	0.05 [0.88]	0.06 [0.93]
Decile	NYSE	VW	0.16 [1.62]	0.14 [1.42]	0.15 [1.47]	0.15 [1.50]	0.16 [1.55]	0.16 [1.58]

Table 3: Conditional sort on size and NIS

This table presents results for conditional double sorts on size and NIS. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on NIS. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high NIS and short stocks with low NIS. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	NIS Quintiles					NIS Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.61 [2.48]	0.74 [3.01]	0.76 [2.89]	0.97 [3.84]	0.90 [3.68]	0.29 [3.90]	0.29 [3.92]	0.28 [3.74]	0.26 [3.45]	0.22 [2.94]	0.21 [2.78]
	(2)	0.73 [3.15]	0.78 [3.42]	0.74 [3.17]	0.88 [3.87]	0.85 [3.70]	0.12 [1.53]	0.13 [1.55]	0.10 [1.20]	0.14 [1.67]	0.11 [1.36]	0.15 [1.75]
	(3)	0.70 [3.22]	0.70 [3.42]	0.76 [3.65]	0.88 [4.30]	0.86 [3.96]	0.17 [2.05]	0.18 [2.12]	0.15 [1.78]	0.12 [1.48]	0.08 [1.01]	0.07 [0.87]
	(4)	0.68 [3.27]	0.68 [3.49]	0.73 [3.90]	0.80 [4.25]	0.78 [3.87]	0.10 [1.24]	0.13 [1.58]	0.09 [1.09]	0.05 [0.60]	0.10 [1.21]	0.07 [0.81]
	(5)	0.48 [2.60]	0.55 [3.34]	0.58 [3.74]	0.57 [3.58]	0.68 [3.91]	0.20 [2.34]	0.24 [2.78]	0.23 [2.64]	0.24 [2.73]	0.18 [2.04]	0.20 [2.22]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	NIS Quintiles					NIS Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	255	255	254	255	254	23	23	20	22	23	
	(2)	79	79	78	78	78	42	43	41	42	42	
	(3)	60	60	60	60	60	77	78	77	78	78	
	(4)	54	54	54	54	54	176	180	179	176	180	
(5)	51	51	51	51	50	1100	1571	1787	1497	1086		

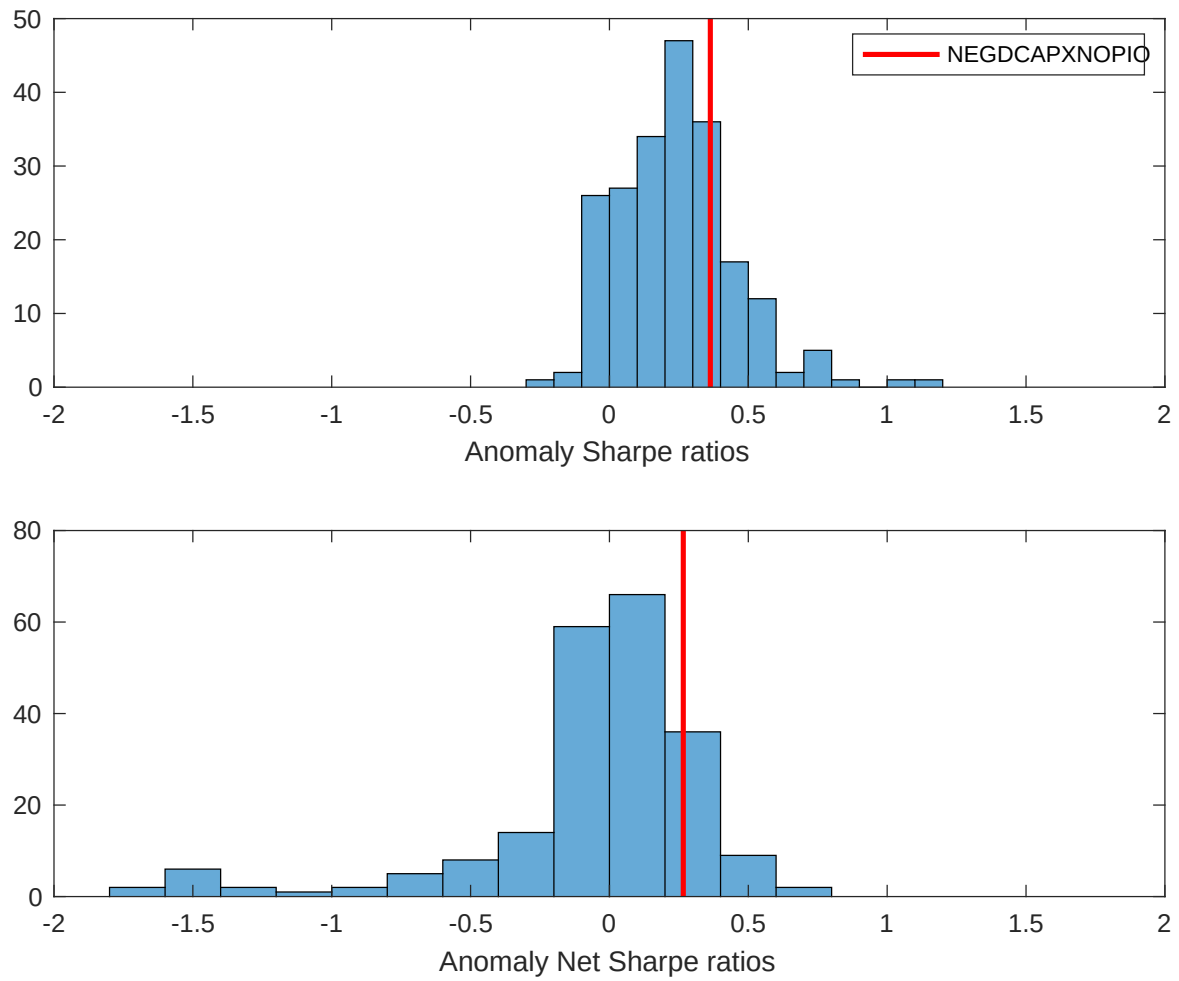


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the NIS with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

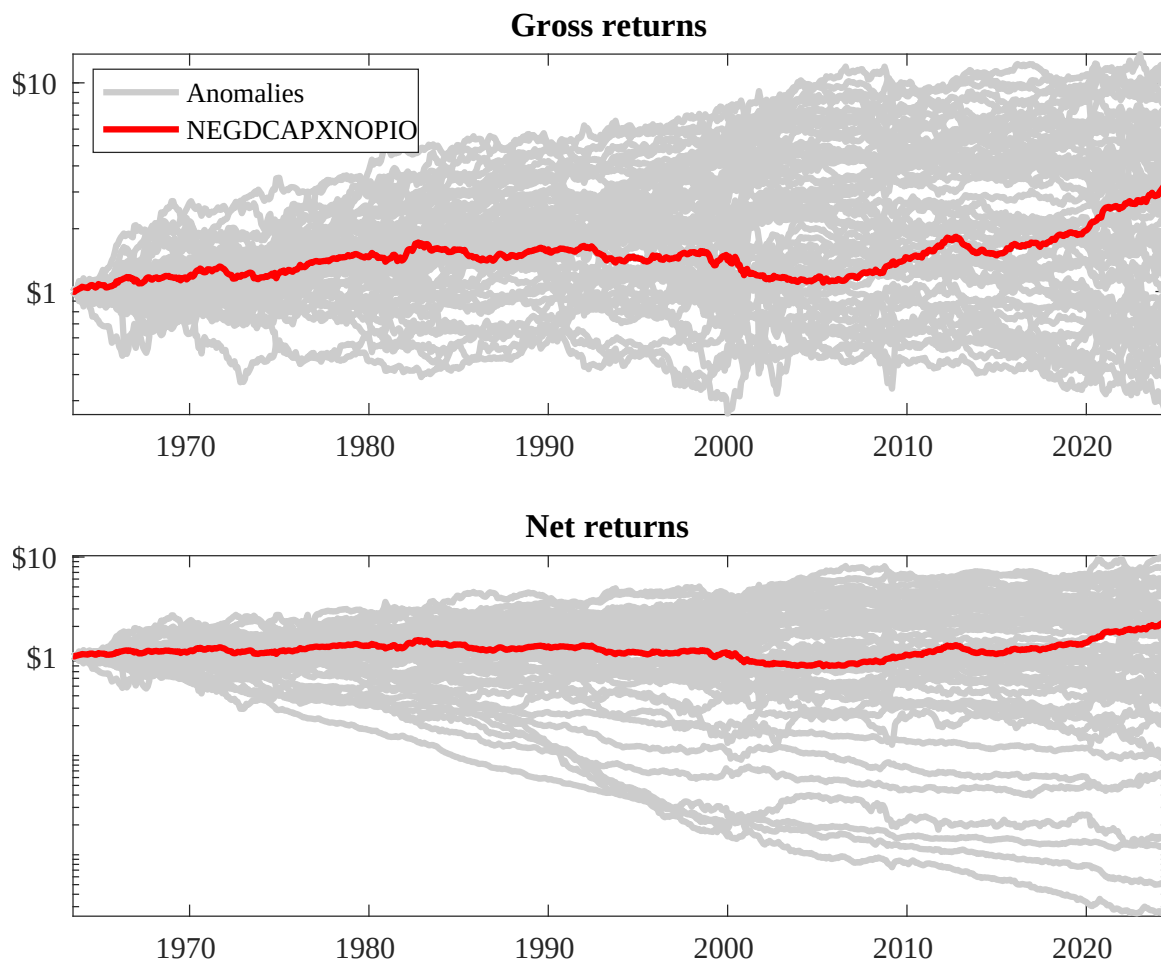
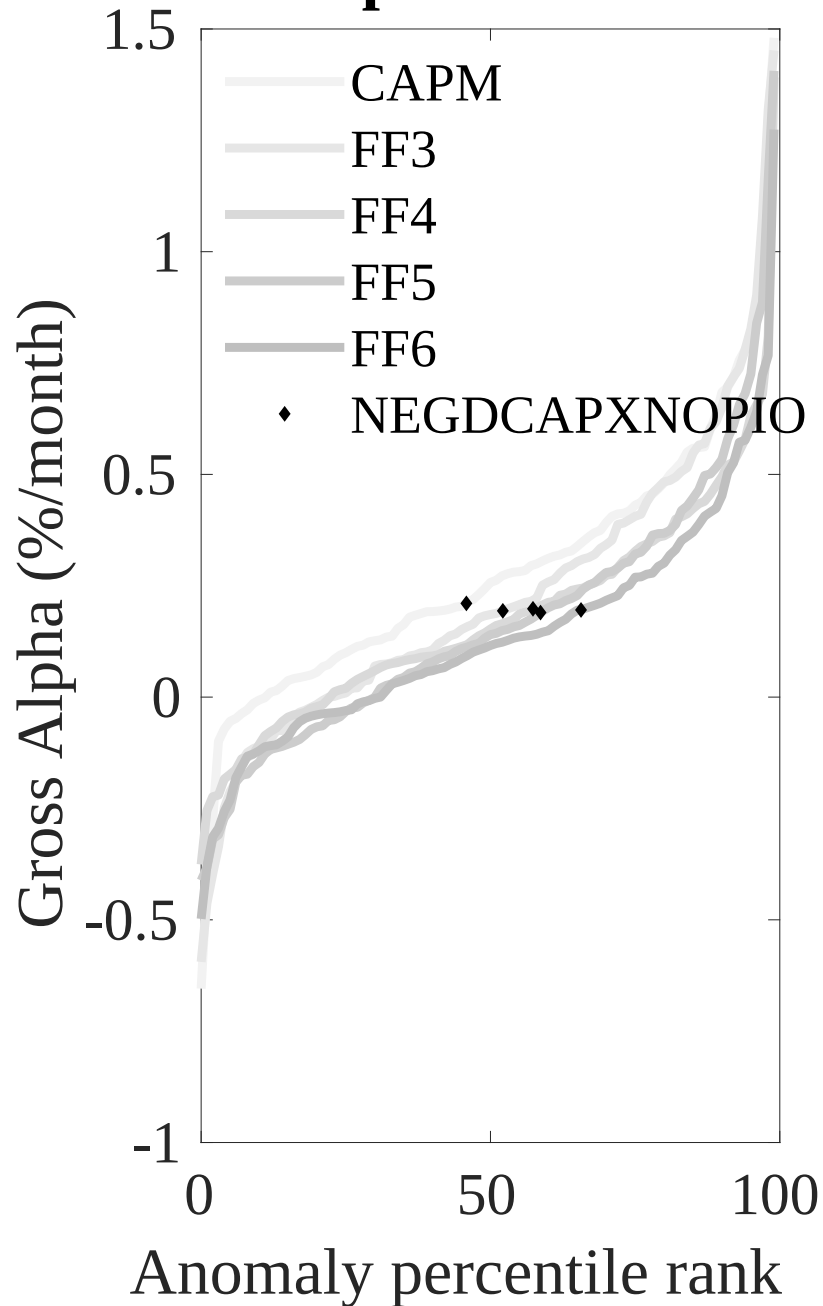


Figure 3: Dollar invested.

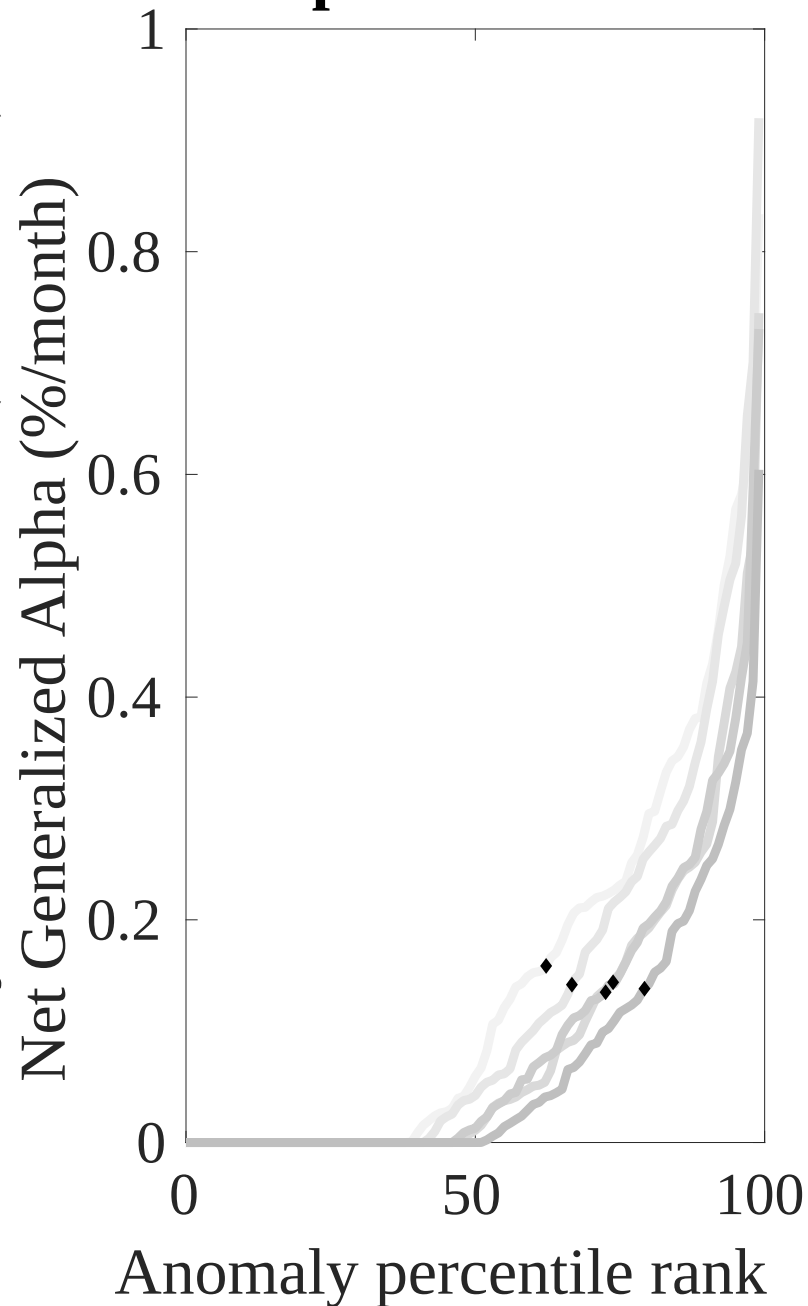
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the NIS trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



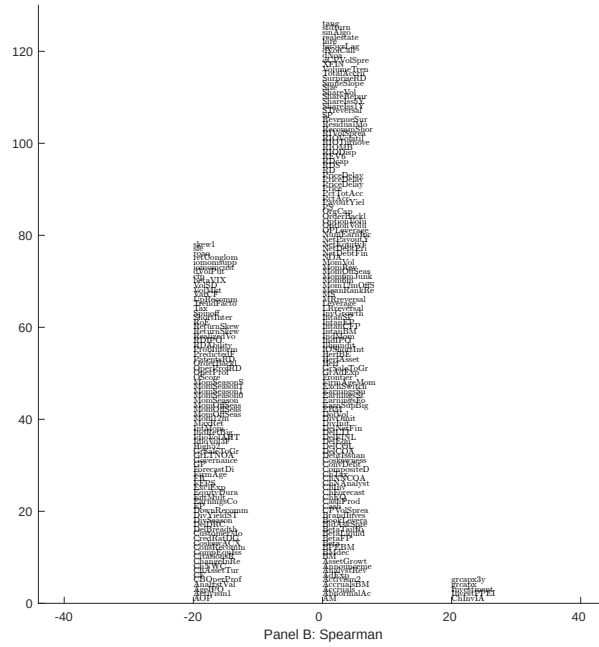
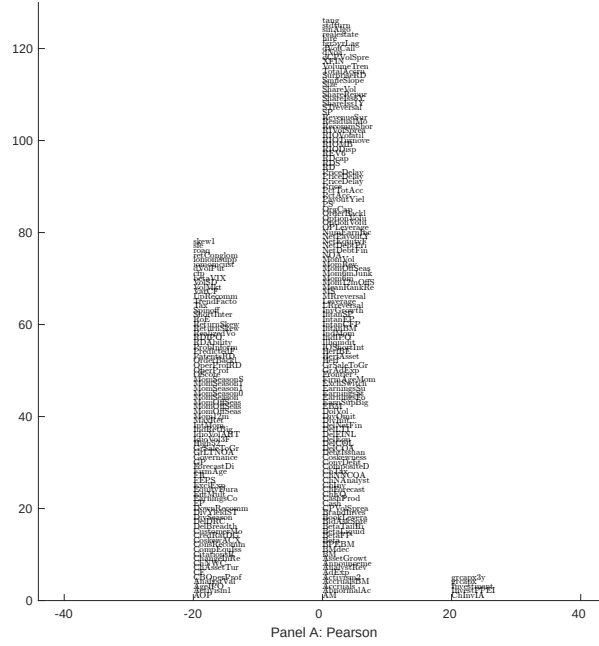


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 209 filtered anomaly signals with NIS. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

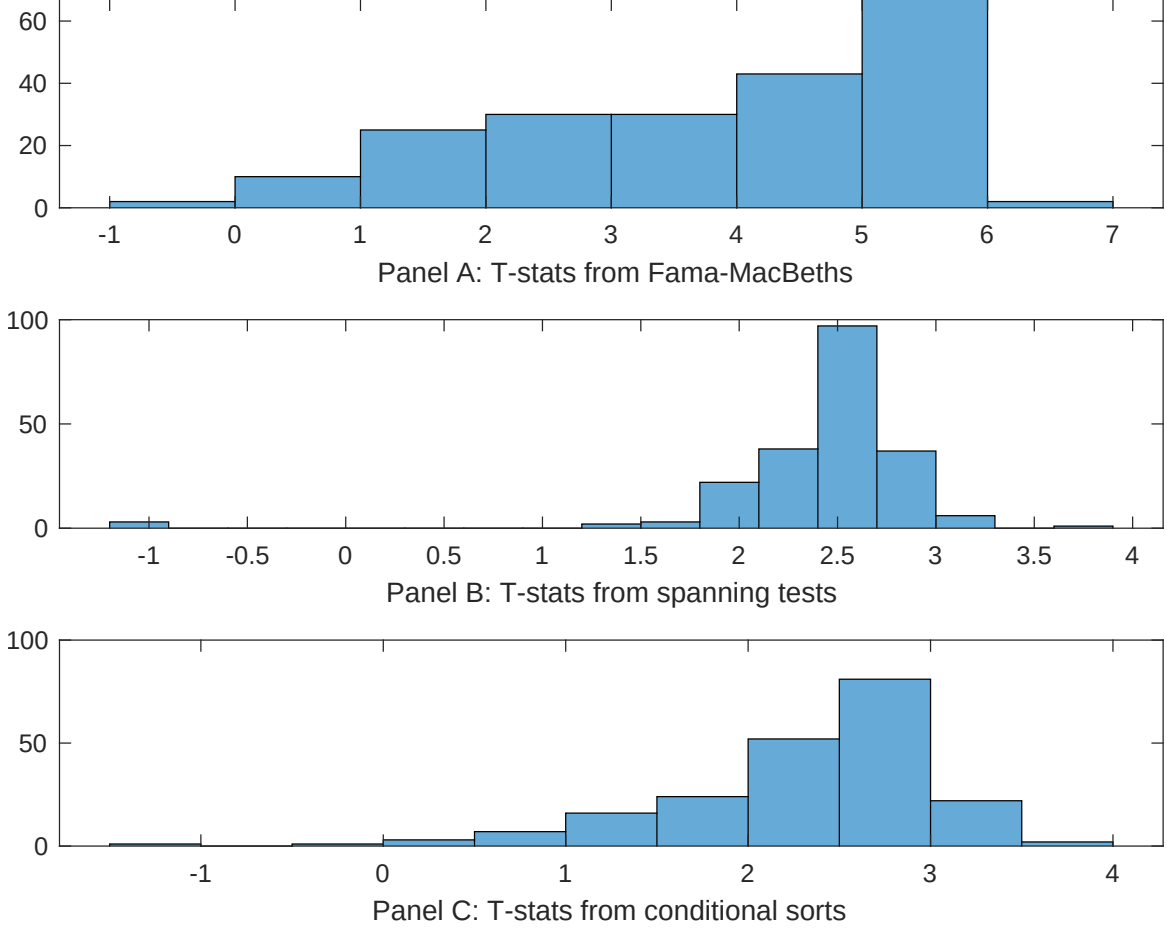


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of NIS conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{NIS} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{NIS}NIS_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{NIS,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on NIS. Stocks are finally grouped into five NIS portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted NIS trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on NIS. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{NIS}NIS_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are change in ppe and inv/assets, Asset growth, change in net operating assets, Growth in book equity, Total accruals, Change in equity to assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.05]	0.13 [6.07]	0.13 [5.85]	0.17 [6.96]	0.12 [5.47]	0.12 [5.57]	0.11 [5.04]
NIS	0.79 [2.80]	0.96 [3.37]	0.11 [3.26]	0.15 [4.70]	0.16 [5.37]	0.16 [4.77]	0.95 [2.88]
Anomaly 1	0.16 [8.09]						0.41 [1.70]
Anomaly 2		0.98 [8.49]					0.38 [1.78]
Anomaly 3			0.14 [9.45]				0.59 [2.39]
Anomaly 4				0.44 [3.94]			-0.23 [-1.42]
Anomaly 5					0.62 [3.04]		-0.88 [-0.33]
Anomaly 6						0.15 [4.27]	0.77 [1.27]
# months	732	732	720	720	732	720	720
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the NIS trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{NIS} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are change in ppe and inv/assets, Asset growth, change in net operating assets, Growth in book equity, Total accruals, Change in equity to assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.20 [2.88]	0.20 [2.89]	0.18 [2.59]	0.19 [2.76]	0.18 [2.57]	0.19 [2.82]	0.19 [2.73]
Anomaly 1	12.92 [4.01]						10.44 [3.06]
Anomaly 2		15.88 [3.87]					15.35 [3.21]
Anomaly 3			11.08 [2.85]				-2.69 [-0.60]
Anomaly 4				6.00 [1.59]			-0.77 [-0.15]
Anomaly 5					15.30 [4.85]		15.88 [4.29]
Anomaly 6						6.54 [1.90]	-8.32 [-1.57]
mkt	-1.65 [-1.01]	-1.48 [-0.90]	-1.44 [-0.87]	-1.30 [-0.78]	-2.02 [-1.24]	-1.61 [-0.97]	-2.04 [-1.25]
smb	-5.06 [-2.13]	-5.82 [-2.43]	-4.42 [-1.85]	-4.74 [-1.98]	-4.48 [-1.90]	-4.62 [-1.93]	-6.03 [-2.54]
hml	-1.94 [-0.61]	-1.08 [-0.34]	-1.40 [-0.44]	-1.25 [-0.39]	-0.42 [-0.13]	-1.39 [-0.43]	-0.70 [-0.22]
rmw	-5.51 [-1.72]	-5.67 [-1.77]	-5.35 [-1.66]	-5.15 [-1.59]	-3.02 [-0.94]	-4.84 [-1.49]	-4.03 [-1.26]
cma	2.93 [0.55]	-6.33 [-0.93]	4.58 [0.83]	7.03 [1.18]	5.16 [1.05]	6.26 [1.07]	-10.55 [-1.49]
umd	-0.75 [-0.46]	-0.24 [-0.15]	-0.96 [-0.58]	-0.83 [-0.50]	-0.37 [-0.23]	-0.57 [-0.35]	-0.04 [-0.02]
# months	732	732	732	732	732	732	732
$\bar{R}^2(\%)$	5	5	4	3	6	3	8

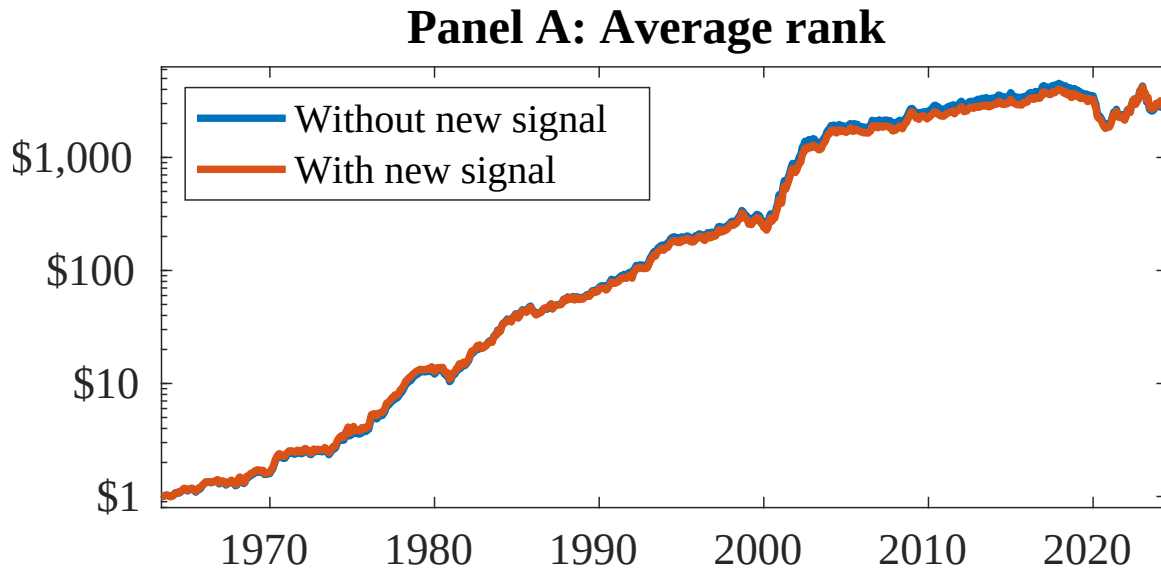


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as NIS. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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